

CSCB07 – Software Design

Project

Logistics

1. This is a group project that is worth 20% of the course grade.
2. The due date is August 6, 2026.
3. Attending the tutorial sessions during the weeks of July 6 and 20 as a group is mandatory and will be used to evaluate your progress and adherence to the requirements.
4. The contribution of each student will be evaluated separately. That is, students belonging to the same group might end up getting different grades.
5. Make sure to review all the material listed under “Useful Resources” on Quercus.
6. Use Piazza for any general questions related to the project.
7. **If you need any technical help, feel free to reach out to your TA, who will schedule an online meeting to address the issue and provide support.**

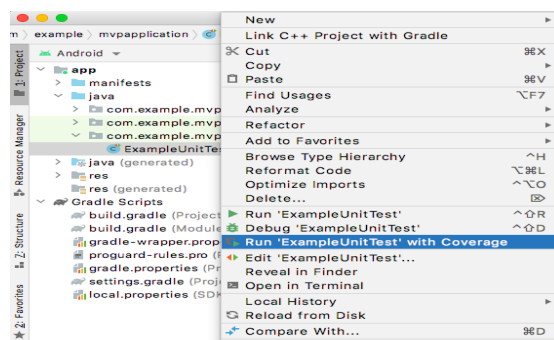
Project Requirements

1. Create a set of user stories based on the description provided in the “Requirements” document and develop an Android application according to these user stories.
2. Use Scrum for the application development process:
 - a. Schedule a meeting as soon as possible to choose a Scrum Master, establish a team agreement, and discuss every aspect in detail: communication methods, expected response times, nature and frequency of meetings during development, contingency plans, etc. This agreement should be signed by all team members and uploaded to Quercus by July 5.
 - b. Conduct “Standup” meetings and document them
 - Each team member should answer the following during the standup meetings:
 - i. What have I completed since the last meeting?
 - ii. What will I finish before the next meeting?
 - iii. Are there any obstacles? If so, what are they?
 - Ideally, these meetings should occur daily, with a minimum requirement of three meetings per week.

- c. You should have 2-3 sprints in total. As such, keep in mind that it's crucial to maintain a balanced workload throughout.
 - d. Collaborate with your team to create a sprint backlog, assigning story points to each task to indicate its complexity. Use Jira to keep track of the user stories and make sure your TA is added to the project. You can refer to this [tutorial](#) for more information regarding Jira usage.
3. Keep track of the changes made to the code using Git and ensure the TA has access to your project as well.
4. Test the application according to the instructions listed below.
5. Record a 5-10 minute video demonstrating the usage of your application.
6. **To be completed individually:** You are required to evaluate the contribution of the other team members through “Peer Evaluation” on Quercus by August 6. The evaluation is based on a scale of 5 and will remain confidential. Failing to submit an evaluation will result in an assumed rating of 5/5.

Testing

1. Refactor the **login/signup module** of your application according to the Model-View-Presenter approach.
2. Write JUnit tests to validate the presenter according to the following guidelines:
 - Each test method should validate a specific behavior and should not contain any code that is not relevant to the behavior being validated. Test methods that violate any of these conditions will be excluded from consideration.
 - Use Mockito to simulate the interactions with the model and the view.
 - Write as many unit tests as needed to maximize line coverage of the presenter. You can generate coverage information in Android Studio as follows:



Submission

The Scrum Master should submit the following deliverables as a single archive file (.rar or .zip) to “Project Deliverables” on Quercus by August 6. Finalizing these deliverables, however, is the responsibility of the group as a whole.

1. The application code
2. Code documentation
3. A document including the following information:
 - a. The main tasks done by each member of the group
 - b. A summary of each of the Standup meetings
 - c. A link to the video

Grading Scheme

1. Group mark M_1 (100 points)
 - Implementing the main functionalities (45 points)
 - User interface design and ease of use (20 points)
 - Clean and cohesive UI design
 - Consistent branding
 - Creative use of design elements
 - User-friendly design with a minimal learning curve
 - Scrum (12 points)
 - Establishing a team agreement
 - Effective and consistent team communication
 - Balanced workload across sprints
 - Proper Jira usage
 - Version control (8 points)
 - Proper use of commits, branching, pull requests, etc.
 - Testing (7 points)
 - Code quality (5 points)
 - Code is easy to read, modify, and re-use
 - Proper documentation

- Video demo (3 points)
2. Individual mark M_2 (10 points)
 - Peer evaluation (5 points)
 - Teaching team evaluation (5 points)
 3. Overall project mark (100 points)

$$Mark = \begin{cases} M_1 & \text{if } M_2 \geq 8 \\ M_1 \times (M_2/8) & \text{otherwise} \end{cases}$$